

Konig theorem

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Claim : For a bipartite graph, the minimum vertex cover equal to the maximum edge matching

Assume that there is k maximum edge match, we observe that since those edge are pairwise no intersect, therefore if we want cover those k edge then we need at least k vertex on those edge

Now we need prove that we only need exactly k vertex is enough, consider a bipartite graph with set A and set B , after finding the maximum edge match we start from the unsaturated vertex in A and walk through the alternating path which is no match edge $\rightarrow$ match edge $\rightarrow \dots \rightarrow$ match edge, it can be shown that the ending edge is always match edge otherwise the original maximum edge match is not maximum, after that we label all vertex we visited in this operation, then now we can choose the vertex which are not visited in A and visited in B , let those chosen vertex form a set S

Prove 1 : $c \in S$ can cover all edges

Consider a edge (u, v) it does not been cover by c then u must been visited in A and v must been not visited in B ,

(u, v) is match edge

if (u, v) is match edge then from the algorithm above v must be visited before u so it can't appear u visited but v not visited

(u, v) is no match edge

if (u, v) is no match edge, according to our algorithm, we will pass through this edge which mean v can't be not visited

Prove 2 : $|S| = k$

Every match edges have only one vertex been chosen in S , from the algorithm above, we observe that for a match edge, either both side vertex been visited or no visited simultaneously so obviously only one side vertex will been chosen. Now we only prove that size of $S \geq k$, now assume that there is at least one vertex $x \in S$ and x doesn't belong to any match vertex, then this will make $|S| > k$ we will prove that it's impossible. Consider x is from A , we choose the not visited vertex in A so it imply that x is saturated but since we assume that x is on

unmatch edge,so x only can from B and also we choose visited vertex in B so it imply that x is saturated also,so contradiction happen so it's impossible exist such x